AREA 1: THERMAL AND FLUID SCIENCES

Overview

Focuses on thermodynamics, fluid mechanics, heat transfer, and energy systems. This domain is critical for modern sustainable energy solutions, electric vehicle optimization, and HVAC systems.

Core & Elective Courses

- **Advanced Heat Transfer:** Study of conduction, convection, and radiation modes. Crucial for designing cooling systems.
- **Thermodynamics & IC Engines:** Fundamental principles of energy conversion and thermal efficiency.
- **Refrigeration and Air Conditioning (RAC):** Design of vapor compression and absorption cycles, cooling load calculations, and eco-friendly refrigerant selection (e.g., R-1234yf).

Key Software & Engineering Tools

- **MATLAB Simscape:** Used for physical modeling and simulating multidomain thermal systems.
- **Ansys Fluent (CFD):** Computational Fluid Dynamics software used to simulate fluid flow, heat transfer, and aerodynamic forces.

Career Paths & Industry Roles

- **Thermal Design Engineer:** Specializes in heat exchanger design, electronics cooling, and industrial thermal management.
- **Battery Thermal Management Systems (BTMS) Specialist:** Focuses on designing energy-efficient cooling and heating systems for electric vehicle (EV) battery packs to prevent thermal runaway and optimize performance.
- **HVAC Engineer:** Designs heating, ventilation, and air conditioning systems for commercial, residential, and industrial applications.

AREA 2: DESIGN AND ROBOTICS

Overview

Focuses on kinematics, dynamics of machinery, computer-aided design, and the integration of electronics with mechanical systems to create smart automated devices.

Core & Elective Courses

- **Computer-Aided Design (CAD):** Principles of 3D solid modeling, parametric design, and geometric dimensioning and tolerancing (GD&T).

- **Finite Element Analysis (FEA):** Stress, strain, and structural deflection analysis under various loading conditions.
- **Mechatronics & Robotics:** Combining mechanical links, sensors, actuators, and microcontrollers to design autonomous systems.

Key Software & Engineering Tools

- **SolidWorks / Autodesk Inventor:** Industry-standard tools for parametric 3D modelling, assembly design, and producing engineering drawings.
- **Ansys Mechanical:** Used for structural FEA simulations, fatigue testing, and vibrational analysis.

Career Paths & Industry Roles

- **CAD/Mechanical Design Engineer:** Responsible for conceptualizing, modeling, and detailing mechanical components and assemblies.
- **Robotics Engineer:** Designs, programs, and maintains robotic arms, automated guided vehicles (AGVs), and kinematic linkages.
- **CAE (Computer-Aided Engineering) Analyst:** Specializes in running structural and structural-dynamic simulations to validate product safety before physical manufacturing.

AREA 3: MANUFACTURING AND AUTOMATION

Overview

Focuses on converting raw materials into finished products using conventional machining, advanced digital manufacturing techniques, and automated production assembly lines.

Core & Elective Courses

- **Additive Manufacturing (AM):** Principles of 3D printing technologies, material selection, and post-processing methods.
- **Design for Additive Manufacturing (DFAM):** Specialized design rules optimization targeting minimal material usage, reducing lightweight structures, and optimizing support removal.
- **Advanced Finishing Processes:** Study of non-traditional surface finish techniques like Abrasive Flow Finishing (AFF) for internal geometries.

Key Software & Engineering Tools

- **Mastercam / Autodesk Fusion 360:** Computer-Aided Manufacturing (CAM) tools used to generate G-code and CNC toolpaths.
- **Ansys Additive Print:** Simulation tool used to predict residual stresses and distortions during metal 3D printing processes.

Career Paths & Industry Roles

- **Manufacturing Engineer:** Optimizes production floor layouts, improves manufacturing cycle times, and oversees tool selection.
- **Additive Manufacturing (AM) Engineer:** Specializes in setting up industrial 3D printers, optimizing print parameters, and implementing DFAM strategies.
- **Automation & Control Engineer:** Designs and PROGRAMS Programmable Logic Controllers (PLCs) and SCADA systems to manage automated factory production lines.